

Preprocessing (one-time)

Warmup Lloyd

T_0 iterations, no freezing

Compute NV

$$NV(s_i) = 1 - \frac{1}{R} \sum \cos(\mathbf{n}_i, \mathbf{n}_j)$$

Assign Tiers

5 tiers from NV thresholds

Compute Thresholds

τ_t, K_c per tier + density

Init State

$c_i = 0, \text{frozen}_i = \text{false}$

Per iteration $t = T_0 + 1 \dots T$

Periodic Refresh (every R iters): query ALL sites

bounds stale-KNN error
~2% amortized overhead

UNFROZEN ONLY | 60-80% COST

KNN Search

compact unfrozen $\rightarrow$ hub-grid
bitonic sort $\rightarrow$ warm-start

FROZEN SITES

Frozen KNN Restoration

copy stored KNN $\rightarrow$ zero cost

ALL SITES

Voronoi Clip $\rightarrow$ Centroid $\rightarrow$ Reprojection

all sites | lightweight

UNFROZEN SITES

Dual-Gate Freeze Test

Gate 1: disp + Gate 2: KNN stable
 $\rightarrow$ streak $\geq \tau_t \rightarrow$ freeze

frozen sites skip KNN | stored KNN reused next iteration